Ammar Nagri

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Profile

BSc Computer Science and Mathematics undergraduate at The University of Manchester, working on theoretical machine learning, neural networks and diffusion models.

Education

The University of Manchester, Computer Science and Mathematics — Manchester, GB
Sept 2025 – June 2028

- Projected First Class Honours.
- Coursework: Probability I, MFA, Statistics I, Data Science, Introduction to Programming I (Python), Introduction to Programming II (Java), First Year Team Project.

Experience

Manchester AI Society, The University of Manchester, Vice President — Manchester, GB
Sept 2025 – present
1 year 3 months

- Vice president running the society's ML workshop programme and its student project track.
- Vice President of the society; previously Project Leader (2025-2026).
- Led workshops teaching machine learning to students, averaging 75 or more students with a 76% retention rate.
- Architected and built an authoritative real-time multiplayer backend for poker using FastAPI and WebSockets.
- Optimised core components to reach 4M or more operations per second in rule evaluation, supporting 40 or more concurrent clients.

Mubit-AI, Software Engineer — London, GB
July 2026 – Sept 2026
3 months

- Software engineer working on the retrieval and storage layer of a memory system.
- Optimised memory retrieval by replacing an O(session size) scan with a near O(limit) reverse scan, achieving up to 2,500x faster limited-history lookups while preserving identical behaviour and chronological ordering.
- Improved storage by implementing tiered per-column-family Zstd compression, reducing disk usage by 35% with minimal latency impact.

Level2, Machine Learning Intern — New York, NY (Remote)
Jan 2026 – Apr 2026
4 months

- Machine learning intern working on regime detection and forecasting for intraday market data.
- Built a Gaussian hidden Markov model classifier for the regime of intra-day market data.
- Developed a live regime forecaster using temporal convolutional networks and PatchTST architectures, using Monte Carlo dropout for Bayesian uncertainty decomposition, achieving an AUC of 76.1% on SPY.
- Presented my findings alongside fellow students, and am currently working on presenting to more financial startups.

JustEat Takeaway (formerly Flyt), Software Engineer — London, GB

Aug 2022 – Aug 2022 (1 month)

- Software engineer working on order analytics microservices in Go.
- Conducted Go benchmarking and pprof CPU and memory profiling to optimise a core microservice, achieving a 4x latency improvement, 3x fewer allocations and a 2x reduction in RSS memory usage.
- Designed and implemented an HTTP endpoint for an order analytics microservice, handling query parsing, SQL database access and robust error handling, defined through OpenAPI specifications and supporting 700,000 daily requests.
- Developed unit tests using httptest and validated functionality through Postman-based integration testing.

Projects

Changes in Stock Market Structure — June 2026 – June 2026

- Stack: Python, pandas, NumPy, scikit-learn, matplotlib, seaborn.
- Built a data pipeline to ingest S&P 500 historical constituents and Polygon market data, filtering prices to each stock's membership window before computing daily return correlation matrices and sector-level heatmaps.
- Modelled evolving market structure with Graphical Lasso sparse partial-correlation networks and Affinity Propagation clustering, visualising annual changes through MDS-based network layouts and animated frames.

Particle Collision Data Analysis with ATLAS — Sept 2023 – June 2024

- Stack: NumPy, pandas, Jupyter, Git, matplotlib.
- Led a team of 5 students to reconstruct the Z0 boson and calculate its invariant mass using NumPy and pandas.
- Attended a masterclass at the University of Oxford, presenting and defending our methodology and findings.
- Co-authored a research poster and delivered a poster presentation at a London physics conference.

Competitions and Awards

International Olympiad of AI, Silver (2025): Awarded Silver on examination performance.

UKMT Mathematics Challenges (2018 to 2024): Multiple Gold Awards across the Junior, Intermediate and Senior challenges, United Kingdom Mathematics Trust.

Technical Skills

Languages: Python, Java, PostgreSQL, MySQL, Golang, R, TypeScript, C++, JavaScript, LaTeX

Libraries: NumPy, pandas, matplotlib, scikit-learn, OpenCV, Hugging Face Transformers

Frameworks and tooling: PyTorch, TensorFlow, FastAPI, Pydantic, Docker, Git, Weights & Biases, AWS, pytest, pprof